

Isokinetic evaluation of anterior cruciate ligament reconstruction: quadriceps tendon *versus* patellar tendon

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Aim. The goal of this prospective randomized study was to compare the isokinetic recovery of thigh strength after anterior cruciate ligament (ACL) reconstruction by using patellar or quadriceps tendon as graft at the 6th month follow-up.

Methods. The authors evaluated 48 patients who underwent arthroscopic ACL reconstruction using patellar tendon (PT group) or quadriceps tendon (QT group) as autograft after a 6 months follow-up undergoing the following tests: the Ergo-jump Bosco System springboard and Universal's FITNET computerized isokinetic system.

Results. The counter movement jump (CMJ) test showed a 24% ($p < 0.01$) strength deficit in patients operated with patellar tendon and 11% in the quadriceps tendon group. Also in the leg press test the greater differences in strength ($p < 0.05$) were verified in the patellar tendon group, above all the peak torque (PT) test carried out at 3 repetitions (15%).

Conclusion. The strength deficit found in the subjects operated with quadriceps tendon were statistically lower in comparison to that verified in the patellar tendon group. A good recovery in thigh strength after 6 months in patients operated with quadriceps tendon could encourage the use of this kind of graft in order to achieve an easier rehabilitation and a faster release of the patient to daily and sports activity.

KEY WORDS: Muscle, skeletal - Anterior cruciate ligament - Rehabilitation - Sports.

Currently, patellar tendon, hamstring tendon or allografts are the grafts of choice for anterior cruciate ligament (ACL) reconstruction. Many authors have studied the isokinetic strength deficits of the

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involved limb after ACL reconstruction with these grafts,¹⁻³ but few are the reports comparing the results obtained with different reconstructive techniques.³⁻⁵

Carter *et al.*⁴ compared the hamstring and quadriceps isokinetic results in patients having autologous patellar tendon or hamstring ACL reconstruction and Lephart⁵ made a comparison between autologous and allograft patellar tendon.

Quadriceps tendon is a graft harvested from the central third of the tendon used with success^{6,7} for ACL-reconstruction especially for the low harvest site morbidity and the biomechanical properties similar to the patellar tendon.⁶

The goal of this prospective randomized study was to compare the isokinetic recovery of thigh strength after ACL reconstruction using patellar or quadriceps tendon as a graft at the 6th month mark which should be the starting point for sports activity after an aggressive rehabilitation program.¹⁻²

Materials and methods

Between 2000 and 2001, a prospective randomized study was carried out on 48 patients (12 female and

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